

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS (NC State, May 2026). Full-stack engineer with production experience in **TypeScript, React, Next.js, and Node.js** -- shipped 10+ features from DB schema to UI at an AI startup. Built **AI/ML integrations across 3 providers and 8+ models** (Claude, GPT-4o, Gemini), real-time event pipelines, and REST APIs on AWS and GCP. Strong in rapid full-stack delivery, code review, and startup-pace ownership.

WORK EXPERIENCE

Isomer AI May 2025 -- Present

Product Engineer Intern

- Owned full backend of a production AI platform -- shipped 10+ customer-facing features using **TypeScript, Node.js, Express, Prisma ORM, PostgreSQL, Redis**, Auth0, Docker, and Sentry; real-time event pipelines and REST APIs serving real users.
- Built **AI/ML integrations across 3 providers and 8+ models** (Claude, GPT-4o, Gemini) via MCP with Temporal.io workflow orchestration, document processing pipelines (PDF gen, email parsing, S3), and Playwright tests -- reviewed PRs across the engineering team.

Mphasis (HP Company) Jun -- Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows (OCR, NLP, ML) improving claim accuracy 25%; automated deployments on **AWS and GCP** with Grafana + Sentry observability.

Capgemini Jun -- Aug 2023

AI/ML Intern

- Kafka + Spark + Hadoop pipelines for real-time data ingestion; cut processing time 15% and latency 20%.

PROJECTS

PulseRank -- Real-Time Leaderboard & Activity Feed 2026

Full-stack platform with **NestJS + Next.js + React** frontend and event-driven backend: MySQL outbox pattern, MongoDB TTL projections, Redis sorted sets for sub-ms reads, Server-Sent Events for live updates. OpenTelemetry + Prometheus observability; GitHub Actions CI/CD.

TypeScript, NestJS, Next.js, React, MySQL, MongoDB, Redis, Docker, Prometheus

AlgoPulse -- Agentic AI Platform 2025

Event-driven **LLM orchestration** platform. Diagnosed API-polling bottleneck; pivoted to Supabase Edge Functions streaming AI reasoning to client. Temporal.io for durable workflow state + Kafka for fault-tolerant coordination -- **2,000 async req/sec** without state loss.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

Glance AI -- Privacy-First RAG System 2025

RAG system for fintech/healthcare with **per-tenant data isolation** at the vector retrieval layer, preventing context leakage and LLM hallucinations. Compliance-ready for regulated data environments with deterministic output architecture.

Python, LLMs, Qdrant, Vector Databases, RAG

EDUCATION

MS Computer Science -- North Carolina State University Aug 2024 -- May 2026

AI, Operating Systems, Data Science, Algorithms, Software Engineering

B.Tech Computer Science -- Mumbai University Aug 2020 -- May 2024

CGPA 9.32/10 | Machine Learning, Big Data Infra, Data Warehousing, Algorithms

SKILLS

Languages: TypeScript, Python, JavaScript, Java, SQL, C

Stack: React, Next.js, NestJS, Node.js, Express, Prisma ORM, Tailwind, REST APIs, Temporal.io, Kafka

AI/ML: OpenAI API, Claude API, Gemini API, LangChain, Hugging Face, PyTorch, RAG, MCP

Infra: AWS, GCP, Docker, PostgreSQL, Redis, MongoDB, Sentry, GitHub Actions, CI/CD

Awards: HackNC24 Top 10 (100+ teams), 6+ national hackathon wins, IEEE Published, Top 1% Sanjeevani (1100+ teams)